EX - 4

# Implement PACKET SNIFFING USING RAW SOCKETS IN PYTHON

AIM: To implement Packet sniffing using raw Sockets in Python and capture network Packet Passing through the network Interface

OBJECTIVE:

To understand how raw sockets work and how network packets are captured and analysis using Python Programming

THEORY:

- Packet Switching is the Process of monitoring and online data Process flowing across a network.
- A Packet Sniffing intercepts Packet and analyzes their contents
- Raw Sockets Provide direct access to lower level network protocols, allowing applications to capture Packets before, they are Processed by the operating System.
- In Python, the Socket module can be used to create raw Sockets for Packet capture System.

In Python, the Socket module can be used to create raw sockets for packet capture.

- This technique is used for network monitoring, troubleshooting, and security analysis

ALGORITHM:

1. Import the required Socket module
2. Create a raw Socket.
3. Bind the Socket to the network interface
4. Receive Packets using the recvfrom() method
5. Display the captured Packet information
6. Repeat the process continuously to monitor network traffic.

PROGRAM:

```
import Socket
Sniffer = Socket.Socket(Socket.AF_INET,
                        Socket.Sock_RAW,
                        Socket.IPPROTO_IP)
host = Socket.gethostbyname(Socket.gethostname())
Sniffer.bind((host, 0))
Sniffer.setSocket(Socket.IPPROTO_IP, Socket.IP_
                                    HDRINCL, 1)
while True:
```

raw_data, addr = ...

print("Packet received from:", addr)

print(raw_data)

OUTPUT:

Packet received from:

('192.168.1.5', 0)

b'E1x001x00 (...

Packet received from:

('192.168.1.10', 0)

b'E1x001x00 (...

RESULT:

Thus, Packet Sniffing has successfully implemented using raw sockets in Python. Displayed packets were compared at displayed